User Manual

1. Install And Run

Install a C++17 compiler, CMake 3.20 or newer, and the system libraries needed by raylib. On Debian/Ubuntu:

```
sudo apt install build-essential cmake libgl1-mesa-dev libx11-dev \
    libxrandr-dev libxinerama-dev libxcursor-dev libxi-dev
```

Build and launch from the repository root:

```
./scripts/build.sh
./scripts/run.sh
```

The checked-in configuration starts windowed at 1280x800 with the CNN activation panel selected. Edit `config.ini` and restart to change startup size, fullscreen mode, default FEN, board palette, model paths, startup architecture, MCTS settings, fonts, or clock settings.

Before a model demo, make sure the runtime model files exist:

```
./scripts/import_models.sh
```

On Windows, use the packaged build in `dist/cpp-nn-visualizer-windows-x64/` and run either `run-cnnv.bat` or `run-cnnv.ps1`.

2. Basic Chess Play

Move input is mouse based. Click a piece and then click a legal destination, or drag a piece to a legal destination. The board highlights the selected piece, legal targets, and the last move. Promotion opens a picker; choose the promoted piece with the mouse or press `Esc` to cancel the pending selection.

The status area reports the game state. The move list records SAN moves and can be clicked to jump to a previous ply. Undo/redo keeps the linked-list move history intact, so rewinding and replaying does not lose future moves unless a new move is made from the rewound position.

3. Command Buttons

Button	Action
Reset	Restores the configured start position and clears transient UI state.
Flip	Reverses board orientation without changing the position.
Undo	Steps one ply backward through move history.
Redo	Steps one ply forward after an undo.
Random	Plays one random legal move from the current position.
Search	Starts or stops the live search preview; the button uses its active color while search is running.
Load FEN	Opens a text dialog for loading a complete six-field FEN.
Save FEN	Writes the current position to `position.fen`.
Setup	Opens the board editor.
Edit FEN	Opens the current position as editable FEN text and then enters setup mode.

4. FEN Loading And Saving

Click Load FEN or press `L` to open the FEN dialog. Enter a complete six-field FEN string and press Enter. Invalid input stays in the dialog and shows an inline error.

Click Save FEN or press `S` to write the current position to `position.fen`. There is no file picker yet.

5. Setup Editor

Click Setup or press `T` to enter editor mode. Select a piece from the palette, left-click a square to place it, or right-click a square to erase it. The palette also includes Eraser, Clear, and Startpos controls.

The editor panel controls side to move, castling rights, en-passant target, halfmove clock, and fullmove number. Click Validate before Apply. Apply is enabled only when the edited position is legal. Invalid positions stay in the editor and show the first validation issues.

6. Activation Views

The right-side architecture selector has NNUE, CNN, BT4, Classical, Off, and a Tree tab. Press `A` to cycle NNUE -> CNN -> BT4 -> Classical -> Off -> NNUE. Changing the architecture does not change the chess position.

The activation panel can be zoomed and panned:

Input	Action
Mouse wheel over panel	Zoom toward the cursor.
Right or middle drag	Pan the panel.
`+` / `-`	Zoom about the panel center.
`0`	Reset panel camera.

NNUE

NNUE shows HalfKP-style active features, raw accumulators, clipped activations, output weights, centipawn value, and learned-weight atlases. Its five internal modes are Overview, Trace, All, Atlas, and Diagram.

CNN

CNN uses the compact `LC0J` runtime file. It encodes the board into 112 planes, runs the residual trunk, policy head, and WDL/value head, and publishes heatmaps and tensor statistics. Its modes are Overview, Trace, All, Atlas, and Diagram.

BT4

BT4 loads the compact `BT4J` v2 runtime file when the view is first selected. The default file is a real native transformer with 64 square tokens, 96-dimensional embeddings, 4 encoder blocks, 4 attention heads, policy logits, and a WDL value head. The view derives block and head counts from the loaded tensors, so it is not tied to one hardcoded transformer size.

BT4 modes are Overview, Trace, All, Atlas, and Diagram. The attention-board views show the current pieces on the board so attention movement can be read in chess terms rather than only as matrix cells.

Classical

Classical is model-free. It evaluates the current position with a handcrafted centipawn evaluator and shows the term breakdown, WDL triplet, phase, and piece-square-table heatmaps. The detailed mode shows the heatmaps by piece type.

Off

Off hides the activation panel and leaves the chess board playable.

7. Search Preview

Press Search or `E` to run a live engine-style preview on the current board. Search stops automatically if the position changes. `Esc` also stops the search. The selected architecture determines the evaluation route when available:

- CNN uses its real WDL head and policy logits for value/priors.
- BT4 uses its WDL head and fallback priors.
- NNUE uses its centipawn value and fallback priors.
- Classical and Off use the handcrafted evaluator.

8. MCTS Tree Workbench

Press `G` or click Tree to open the full-window search-tree workbench. Click Start to run PUCT search from the current position. Click Stop to stop it, Fit to frame the tree, and Board or `Esc`/`G` to return to the main board.

Tree controls:

Control	Action
visits +/-	Change the playout budget.
cpuct +/-	Change the PUCT exploration constant.
Follow	Trace the currently explored leaf into the activation views.
branches +/-	Limit children shown per node; 0 shows all.
depth +/-	Limit displayed tree depth.
Batch	Collapse sibling leaf groups into summary blobs.
Guides	Toggle per-ply guide lines.
Merge	Merge repeated position signatures for transposition display.

Tree navigation:

Input	Action
Mouse wheel	Zoom.
Left drag	Pan.
Click node	Select that node and trace its FEN into the network views.
Space	Play/pause recorded growth frames.
Left / Right	Step the growth scrubber.
Home / End	Jump to first growth frame / live tree.

9. Keyboard Shortcuts

Shortcut	Action
`F`	Flip board.
`R`	Reset to configured start position.
`L`	Open Load FEN.

Shortcut	Action
`S`	Save current position to `position.fen`.
`T`	Enter setup editor.
`A`	Cycle activation architecture.
`E`	Start/stop search preview.
`G`	Enter/leave tree workbench.
`F11`	Toggle borderless fullscreen.
`Ctrl+Z`	Undo one ply.
`Ctrl+Y`	Redo one ply.
`Left` / `Right`	Step backward / forward one ply in board mode; scrub tree frames in tree mode.
`Home` / `End`	Jump to start/end of move history in board mode; first/live frame in tree mode.
`Esc`	Deselect, close modal, stop search, leave editor/tree, or cancel transient input.

10. Configuration

Important `config.ini` keys:

```
window.width = 1280
window.height = 800
window.fullscreen = false
default.fen = rnbqkbnr/pppppppp/8/8/8/8/PPPPPPPP/RNBQKBNR w KQkq - 0 1
assets.pieces = assets/pieces
board.palette = red
startup.arch = cnn
startup.tree = false
mcts.visits = 20000
mcts.cpuct = 2.8
model.nnue = models/nnue-halfkp-demo.bin
model.lc0_cnn = models/lc0-cnn-small-112p-4x32-policy4672-wdl3.bin
model.lc0_bt4 = models/lc0-bt4-tiny-96x4x4h.bin
clock.enabled = false
```

`startup.arch` accepts `nnue`, `cnn`, `bt4`, `classical`, or `off`. `board.palette` accepts `red`, `classic`, `warm`, `neural`, `analysis`, `heatmap`, `blue`, `cool`, `green`, or `forest`.

11. Restrictions

- Move input is mouse-based; there is no SAN/UCI move-entry box.
- FEN loading expects a complete standard six-field FEN string.
- PGN export exists in the codebase, but PGN import is not implemented.
- There is no engine-opponent play mode yet.
- The bundled CNN and BT4 models are compact untrained visualization models. They demonstrate the real native architecture paths, not upstream engine strength.
- The app reads local compact binary model formats (`NNUE`, `LC0J`, `BT4J` and legacy `BT4V` fallback metadata), not official LC0 protobuf files.
- Save FEN writes to `position.fen`; there is no file picker yet.

12. Troubleshooting

- If pieces are missing, check `assets.pieces` and run from the repository root or package directory.
- If model panels report missing weights, check `model.*` paths and rerun `./scripts/import_models.sh`.

- If text input in the FEN dialog is wrong, switch the active keyboard/input method to plain Latin text.
- If the window opens too large or too small, edit `window.width`, `window.height`, or `window.fullscreen` in `config.ini`.